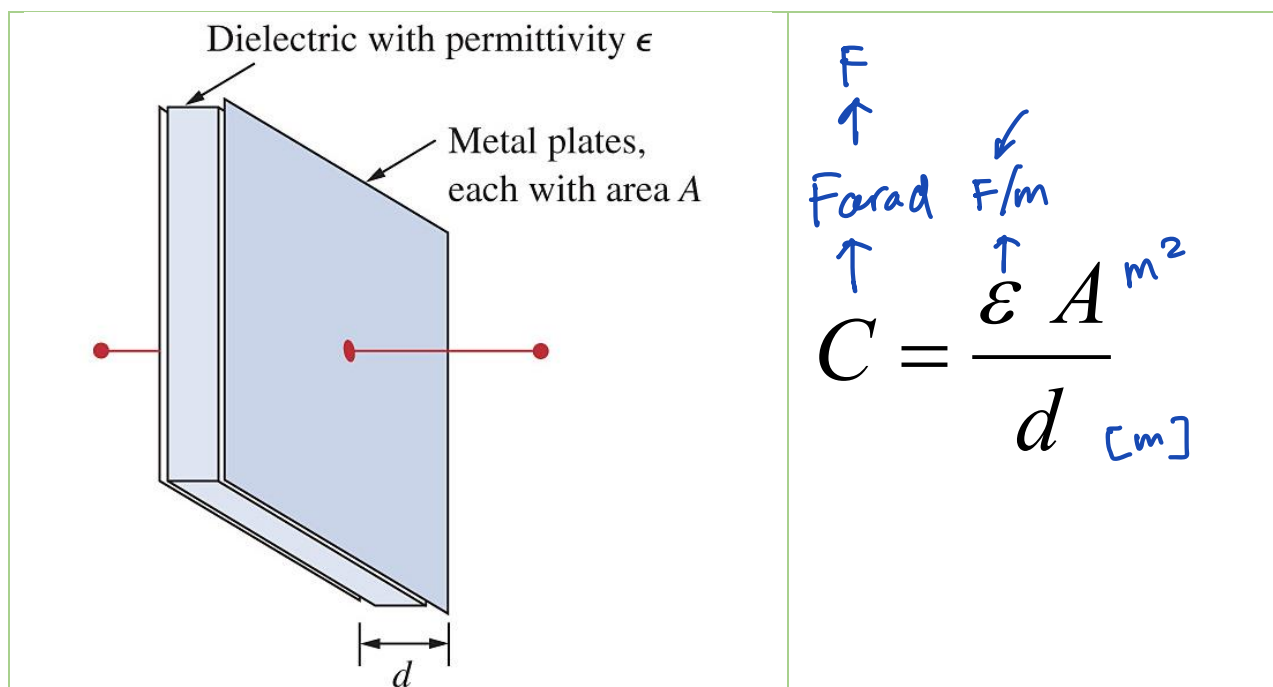


L#22: Capacitor

Physical characteristics

- A capacitor is a passive element designed to store energy in its electric field.
- A **capacitor** consists of 2 conducting plates separated by insulation (dielectric).



A :

d :

ϵ :

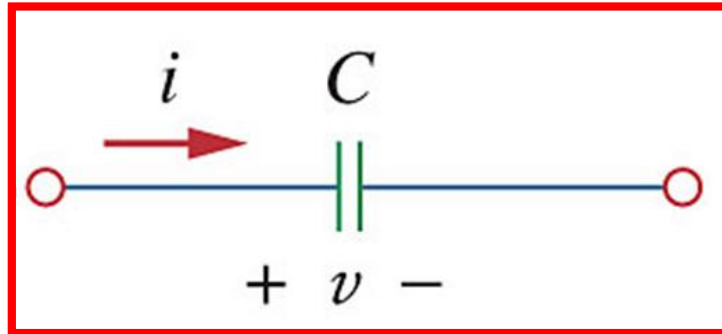
ϵ_r :

ϵ_0 : $8.854 \times 10^{-12} \text{ F/m}$

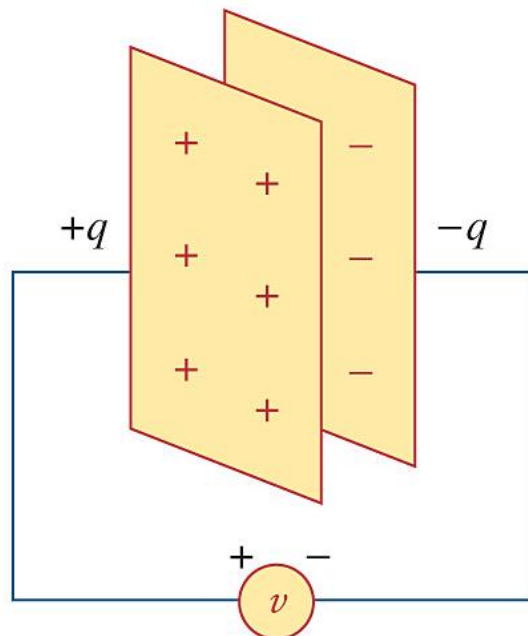
$8.854 \times 10^{-12} \text{ F/m}$

Electrical characteristics

- Capacitance (C): measured in Farad (F).



passive sign convention applies.



$$C = \frac{q}{v}$$

$$q = Cv$$

Current:

$$i = \frac{dq}{dt} = \frac{dCv}{dt} = C \frac{dv}{dt}, \quad i(t) = C \frac{dv(t)}{dt}$$

Voltage:

$$\int_{-\infty}^t i(t) dt = \int_{-\infty}^t C \frac{dv(t)}{dt} dt = Cv(t)$$

$v(t_0)$ is given.

$$v(t) = \frac{1}{C} \int_{-\infty}^t i(t) dt$$

t_0 . ZV.

$$= \frac{1}{C} \int_{-\infty}^{t_0} i(t) dt + \frac{1}{C} \int_{t_0}^t i(t) dt$$

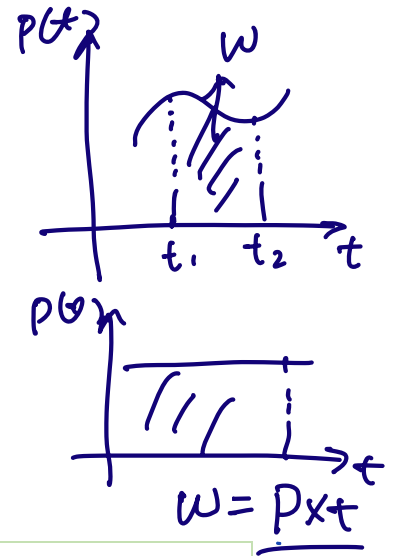
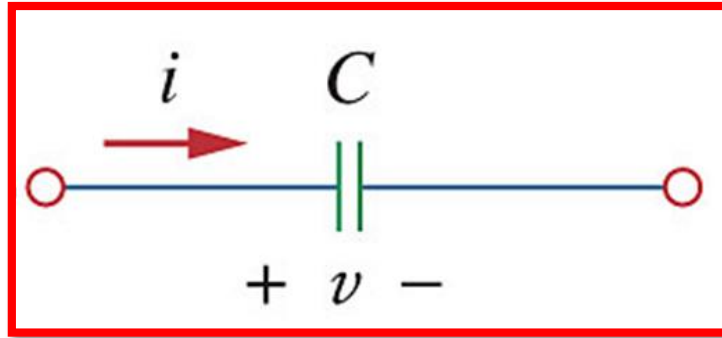
$$\underline{\underline{v(t) = v(t_0) + \frac{1}{C} \int_{t_0}^t i(t) dt}}$$

Energy and Power of Capacitors

$$\int x dx = \frac{1}{2} x^2$$

$$\left(\frac{1}{2} x^2\right)' = x$$

$\downarrow$
 $2x$



Energy:

$$w = \int_{-\infty}^t p(t) dt = \int_{-\infty}^t C v \frac{dv}{dt} \cdot dt = \int_{v(-\infty)}^{v(t)} C v dv = \frac{1}{2} C v^2$$

Power:

$$p(t) = v i = v \cdot C \frac{dv}{dt} = C v \frac{dv}{dt}$$

Important characteristics of capacitors:

- It is an open circuit for DC. $\frac{dv}{dt} \rightarrow 0$.

$$i = C \frac{dv}{dt} \rightarrow 0$$

- Its voltage $\uparrow \infty$ does not change abruptly.

- It does not dissipate energy but only store energy when charging or release energy when discharging.

Examples #1-2

V ?

C.

#1: What is the voltage across a **3 μF** capacitor if the charge on one plate is **0.12 mC** . How much energy is stored in the capacitor?

$$q = CV.$$

$$V = \frac{q}{C} = \frac{0.12 \times 10^{-3} \text{ C}}{3 \times 10^{-6} \text{ F}} = 40 \text{ V.}$$

$$\begin{aligned} W &= \frac{1}{2} CV^2 = \frac{1}{2} \times 3 \times 10^{-6} \times 40^2 \\ &= \frac{1}{2} \times 3 \times 16 \times 10^2 \times 10^{-6} \\ &= 24 \times 10^{-4} [\text{J}]. \\ &= 2.4 \text{ mJ.} \end{aligned}$$

#2: The voltage source connected across a **10 μF** capacitor is given as:

$v(t) = 50 \sin 2000t \text{ V}$. Calculate **$i(t)$** .

$$\begin{aligned} i(t) &= C \frac{dv}{dt} = 10 \times 10^{-6} [50 \sin 2000t]' \\ &= 10 \times 10^{-6} \times 50 \cos 2000t \times 2000 \\ &= \cos 2000t [\text{A}]. \end{aligned}$$

$$i(t=3\text{s}) \rightarrow$$

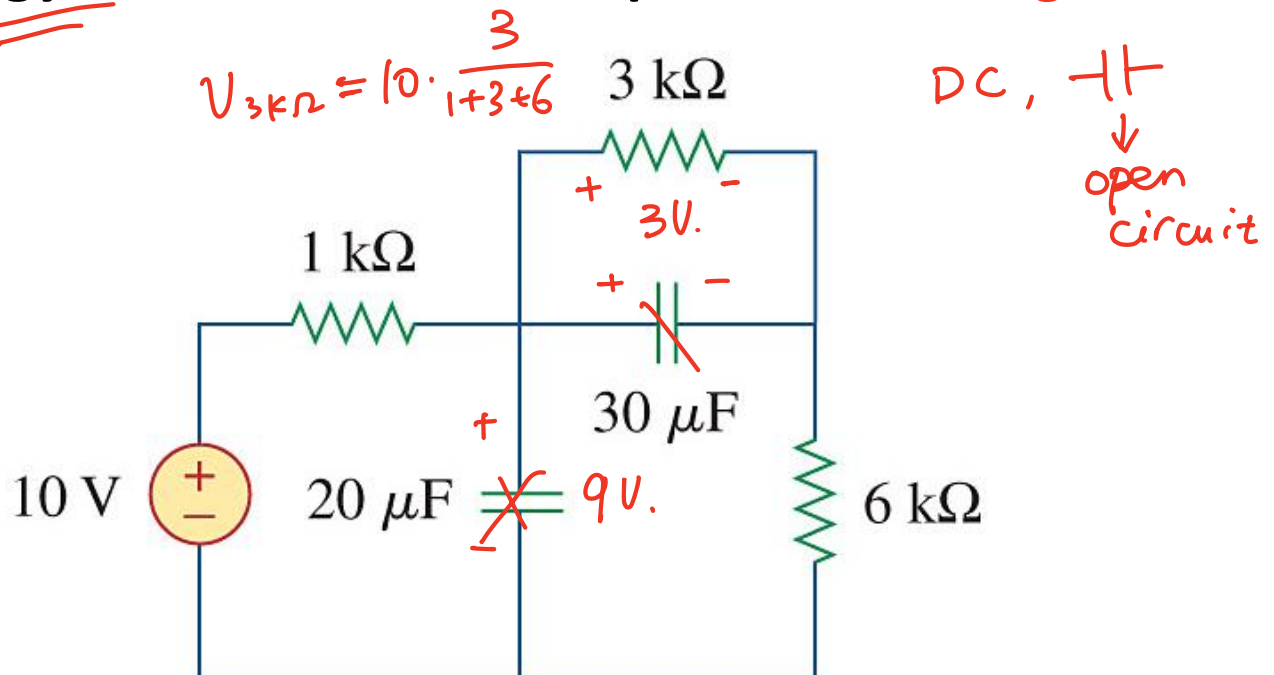
Example #3: If the current through a $100\ \mu\text{F}$ capacitor is given as:

93.14 mV

$i(t) = 50 \sin(120\pi t)$ mA, calculate $v(1\text{ms})$ and $v(5\text{ms})$. Assume $v(0) = 0\text{ V}$

$$v(t) = \int$$

Example #4: For DC conditions, calculate the energy stored in the two capacitors. $w = \frac{1}{2} CV^2$.

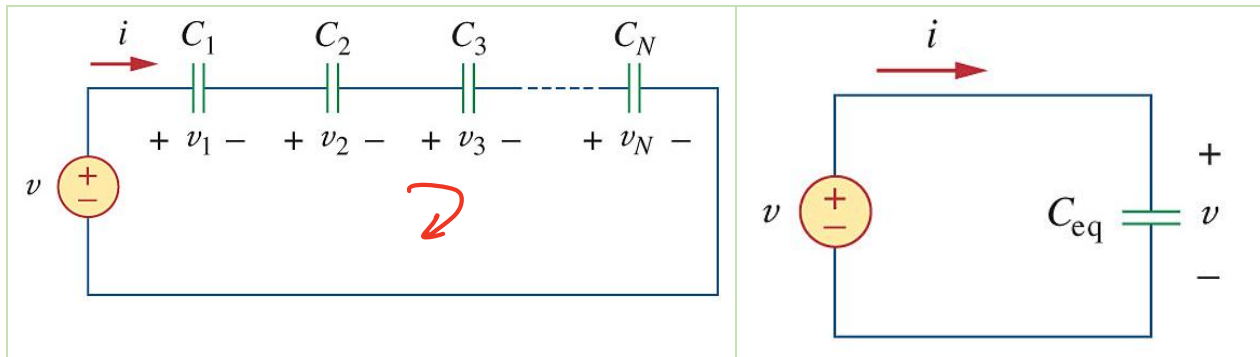


$$W_{30\mu F} = \frac{1}{2} \times 30 \times 10^{-6} \times 3^2 = \text{--- [J]}.$$

$$W_{20\mu F} = \frac{1}{2} \times 20 \times 10^{-6} \times 9^2 = \text{--- [J]}.$$

$$C = \epsilon \frac{A}{d}$$

Series and Parallel Capacitors



$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \dots + \frac{1}{C_N}$$

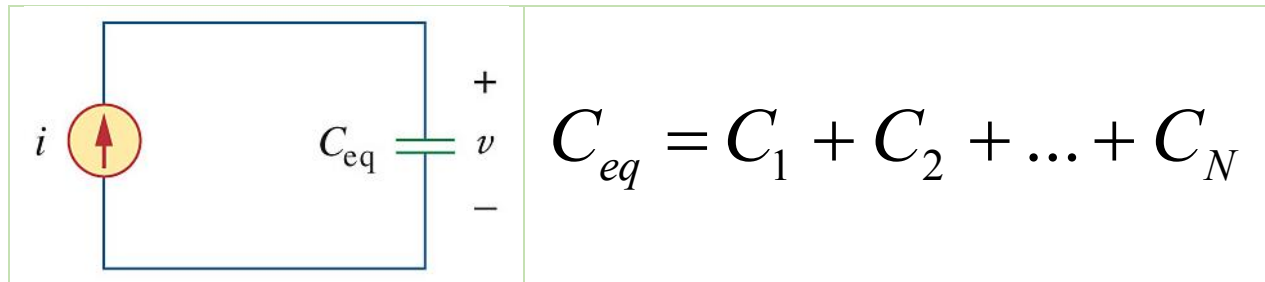
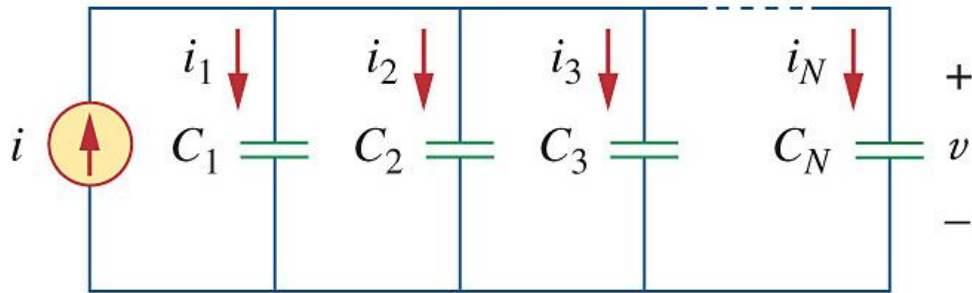
$$v = v_1 + v_2 + \dots + v_N.$$

$$\frac{1}{C_{eq}} \int_{-\infty}^t i(t) dt = \frac{1}{C_1} \int_{-\infty}^t i_1(t) dt + \dots + \frac{1}{C_N} \int_{-\infty}^t i_N(t) dt.$$

$$i_1 = i_2 = \dots = i_N = i(t)$$

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \dots + \frac{1}{C_N}.$$

$$C = \epsilon \frac{A}{d}$$



$$i = i_1 + i_2 + \dots + i_N$$

$$C_{eq} \frac{dv}{dt} = C_1 \frac{dv_1}{dt} + C_2 \frac{dv_2}{dt} + \dots + C_N \frac{dv_N}{dt}$$

$$C_{eq} = C_1 + \dots + C_N$$

Example #5: Determine the equivalent capacitance.

